2-2. 偏振（振动方向） $(-\frac{1}{2}, -\frac{1}{2}, 1)$.

行进方向 $\rightarrow k$. $(1,1,1)$.

相位速度 $v_p = \frac{\omega}{k} = \frac{3\times10^4}{10} = 3\times10^3$

振幅 $\frac{\sqrt{6}}{2}$

频率 3×10^4

波长. $\frac{2\pi}{k} = \frac{1}{10\sqrt{3}} = \frac{\sqrt{3}}{30}$

2.5. $(1, 1.5)$. $\lambda = 500nm = \frac{2\pi}{k} \Rightarrow k = \frac{2\pi}{500nm}$

$= 10^{-6} \cdot 4\pi \times 10^{3}$

$k_z = \frac{\sqrt{3}}{2}$

传播方向 $(1, \frac{\sqrt{3}}{2}, \frac{\sqrt{3}}{2})$.

$f = 2\times10^3 mm^{-1} = \frac{k}{2\pi}$

空间圆频率 $4\pi\times10^3 mm^{-1}$

$kdr - \omega dt = 0$

$kr - \omega t = 0$ $\frac{dx}{dt} = \frac{\omega}{k} = v$

2-12. $v_g = \frac{dk}{d\omega}$ $\omega = kv = k\cdot\frac{c}{n}$ $kc = n\omega$

$cdk = ndw + \omega dn$

$v_g = \frac{d\omega}{dk} = \sim$

2-13.

(1) $\frac{c}{v_g} = n - \lambda \frac{dn}{d\lambda}$ ✡ ···

(2) $v_g = \frac{d\omega}{dk} = \cdots$

(3) $v_p = \frac{c}{n}$. ∴ $n = \frac{c}{v_p}$.

$$\frac{c}{v_g} = n - \lambda \frac{dn}{d\lambda} = \frac{c}{v_p} - \lambda \frac{c d\frac{1}{v_p}}{d\lambda} = \frac{c}{v_p} + \lambda \cdot \frac{c\, dv}{v_p^2\, d\lambda}$$